

Are Viral Exposures Inducing Accelerated Ageing?

Matt McElheron^{1,2,3}, Jordy Smith¹, Darragh Duffy², Etienne Patin³, Lluís Quintana-Murci³, Nollaig Bourke¹

¹ Discipline of Medical Gerontology, School of Medicine, Trinity Translational Medicine Institute, Trinity College Dublin, Dublin, Ireland.

² Translational Immunology Unit, Department of Immunology, Institut Pasteur, Université Paris Cité, Paris, France.

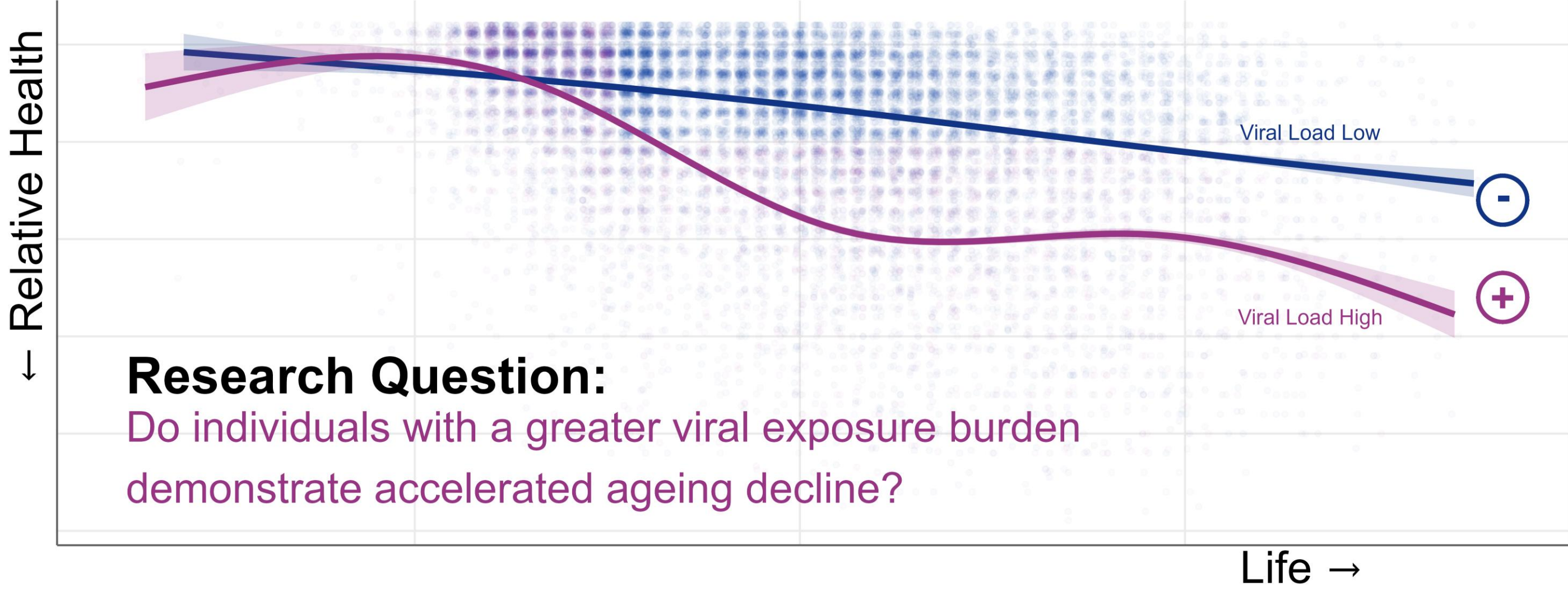
³ Human Evolutionary Genetics Unit, Institut Pasteur, Université Paris Cité, CNRS UMR2000, Paris, France.

1.

Research Background

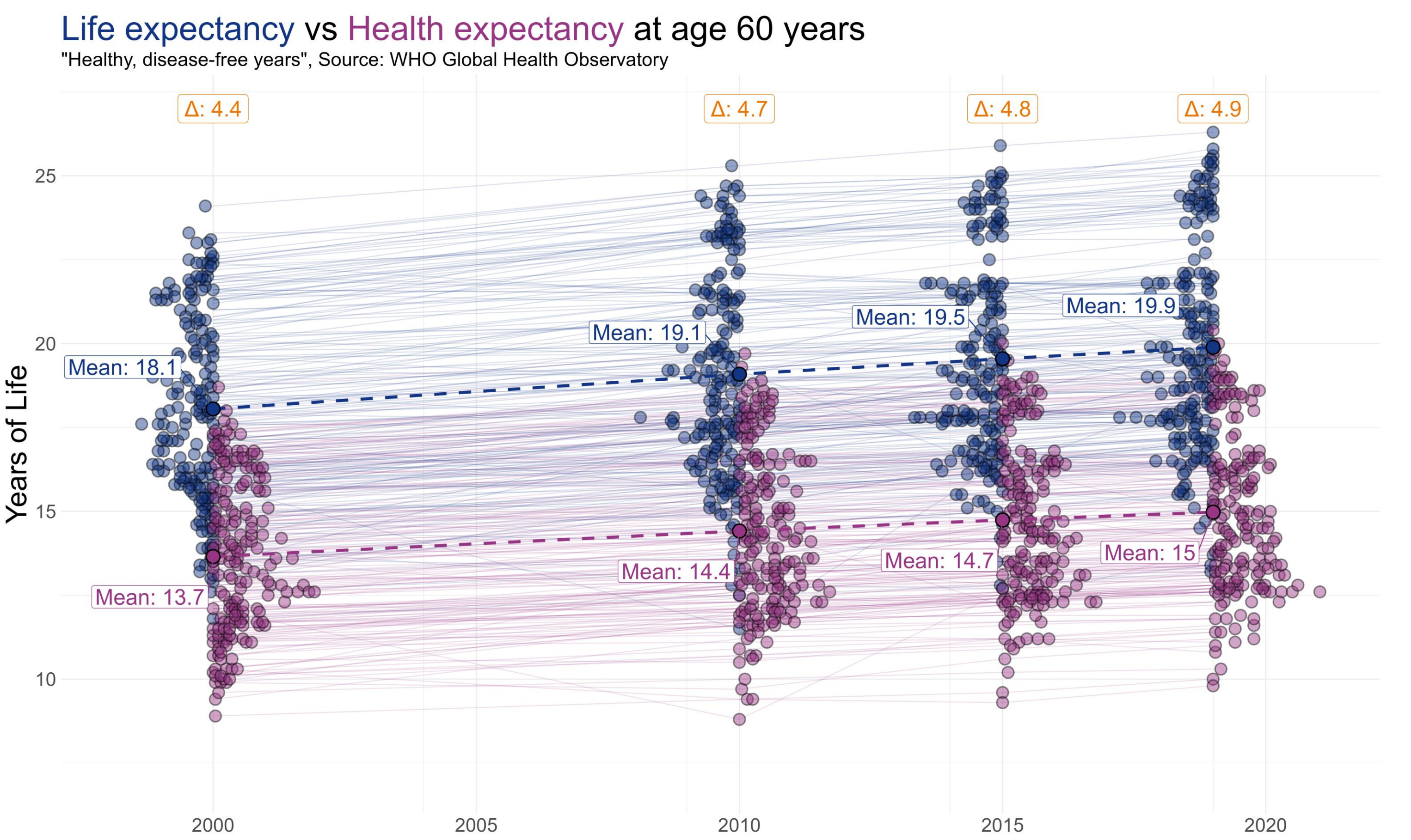
Global Ageing Population: Individuals aged 60+ is projected to double by 2050, from 1 billion to 2.1 billion.
That's good news.

The Healthspan-Lifespan Problem: Divergence between biological and chronological ageing is widening, meaning larger proportions of later life in poor health and straining healthcare systems.
That's bad news.



Persistent Viral Drivers: Persistent viral infections drive chronic inflammation and T-cell exhaustion, but their long-term impact on differential ageing trajectories remains unclear.

Core Hypothesis: Viral exposures accelerate biological and immune ageing in an age-dependent manner, contributing towards later life morbidity burden.



2.

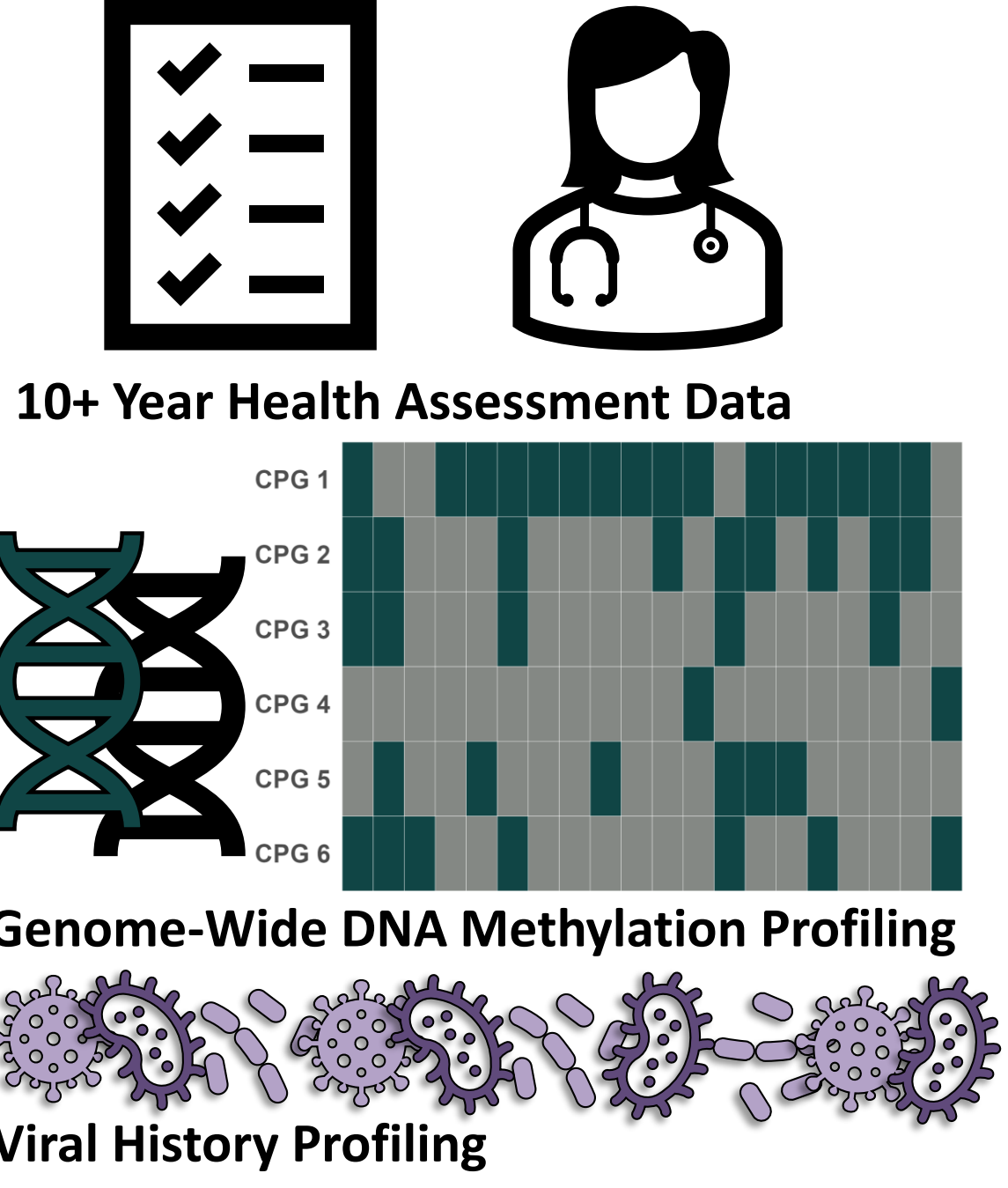
Methods

The Milieu Intérieur Cohort

The *Milieu Intérieur* cohort is a deeply phenotyped, population-based study of 1,000 healthy individuals aged 20 to 69 (stratified by age and sex) living in France.

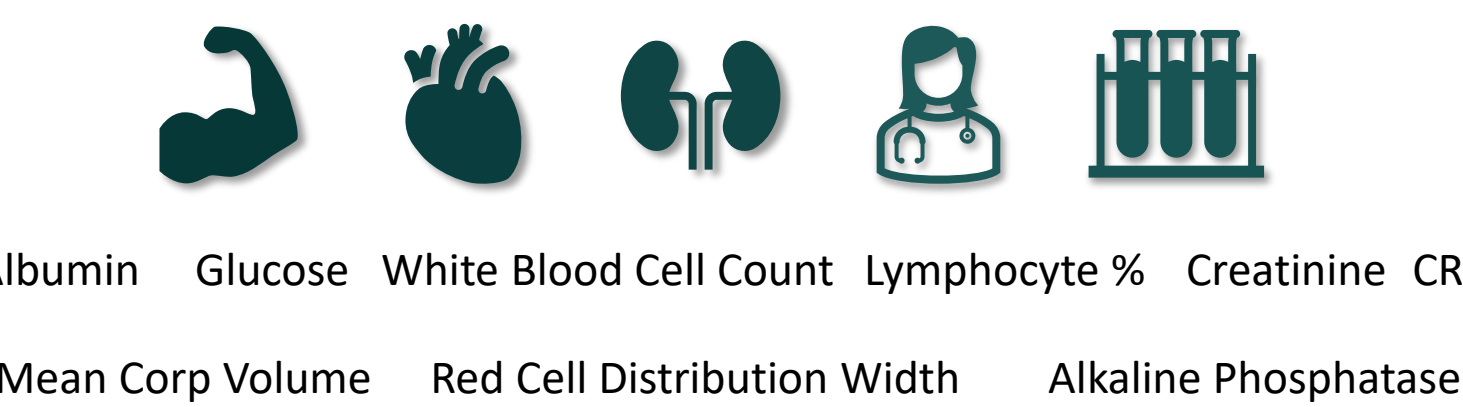
Milieu Intérieur's goal is to establish the boundaries of a healthy immune response and identify the genetic, epigenetic, and environmental factors – including chronic viral exposure – that drive inter-individual variability and immune decline across the lifespan.

N = 1,000
500 x Healthy Men & Women



Immune Cell Ageing Model

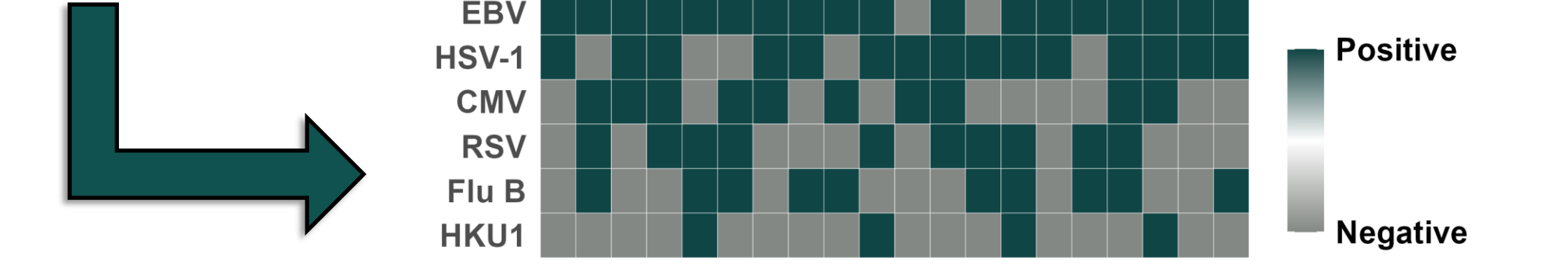
DNAm PhenoAge incorporates 513 CpG sites selected via elastic net regression to surrogate phenotypic age. The underlying phenotype was constructed from mortality risk models trained on 9 clinical biomarkers:



$$DNAm\ PhenoAge = w_0 + \sum_{i=1}^{513} w_i \cdot CpG_i$$

Viral Sero-Epidemiology

- 208 distinct Luminex magnetic bead regions were activated and coupled with viral antigens via covalent amide bonds.
- Diluted serum (1/200 final dilution) was incubated with the pooled antigen-coupled beads in a 96-well format to capture pathogen-specific IgG antibodies.
- Bound IgG was tagged using an R-phycoerythrin-conjugated secondary antibody, and median fluorescence intensity (MFI) was read on the Luminex INTELLIFLEX® system.
- Seropositivity was classified using Bayesian Gaussian Mixture Models or population percentiles.

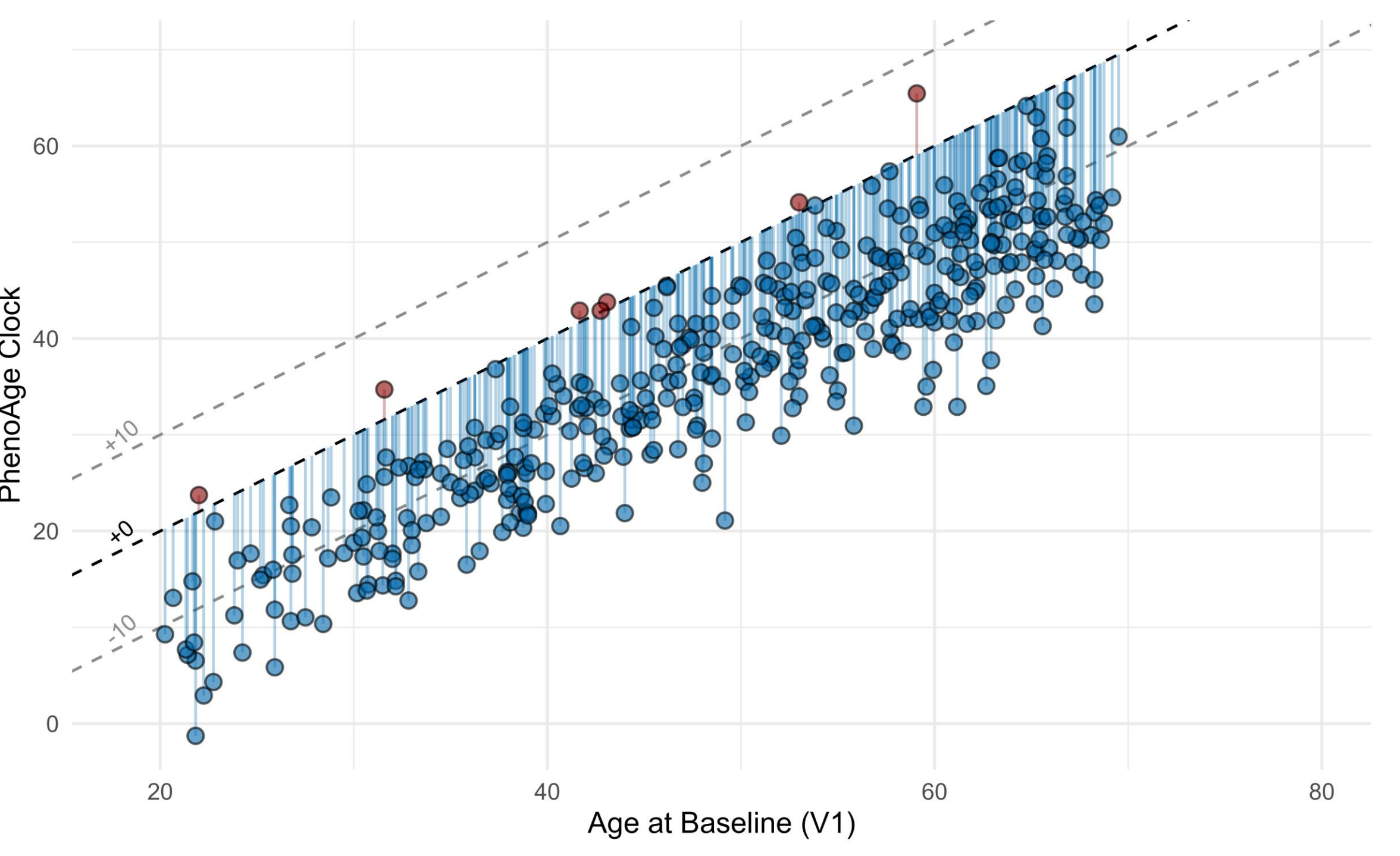


3.

Results

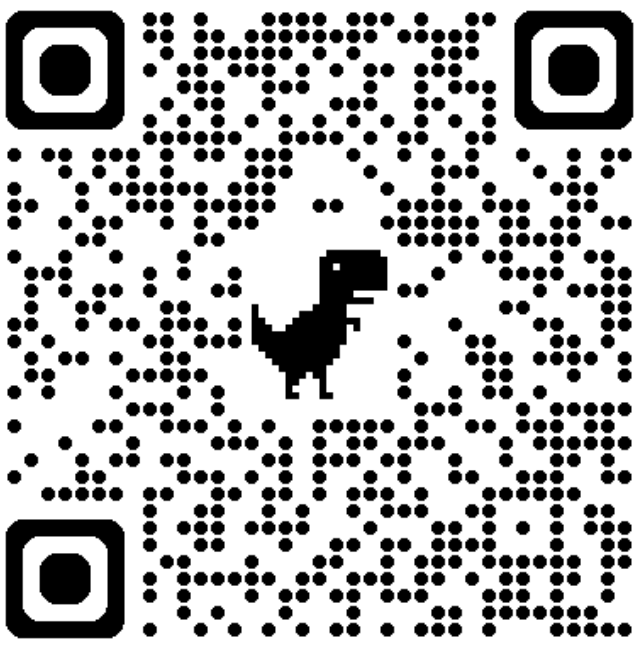
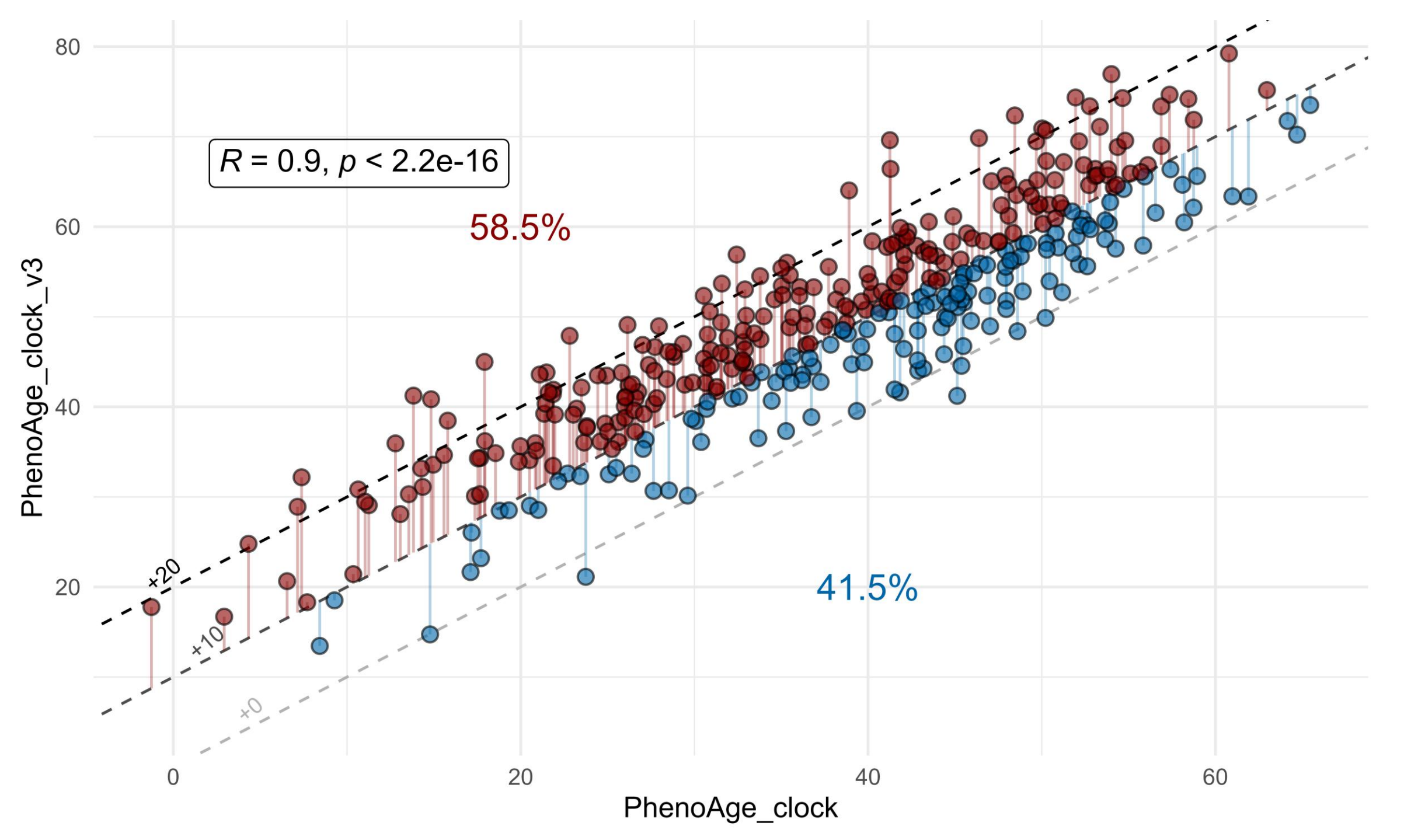
Milieu Intérieur Cohort Immune Methylomes Appear Around 10 Years Biologically Younger Than Expected

● Ageing Slower ● Ageing Faster



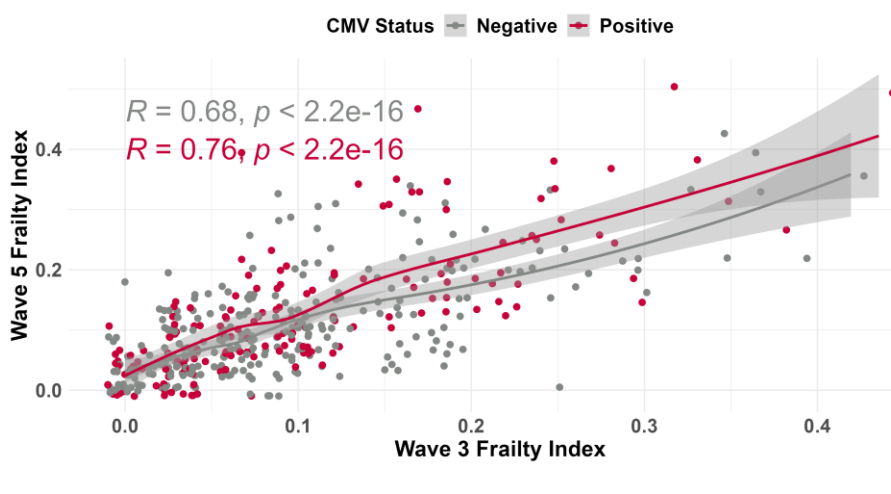
After 10 years, some individuals have aged more than others, with the population average being around +10

● Ageing Slower ● Ageing Faster



github.com/MattMcElheron/McElheron_ISI_Poster

Follow the QR code or link to see figure animations, further visualizations, explanations of caveats, and subsequent findings using a nationally representative cohort.



10-year age acceleration is strongly associated with compositional changes, not viral exposures, in healthy individuals

● Ageing Slower ● Ageing Faster

